

Basis-independent EndRing trapdoor-interface funnel

GOAL-SSI-2fa82a / BATCH-cb7b9a

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Result

This batch ran the requested **propose -> evaluate -> gap-feed -> propose -> evaluate** method at the interface layer that blocked the preceding EndRing PKE candidates. It generated three typed **Trap_E** packets, froze both review inputs, and obtained independently owned T1–T5 verdicts. The result is **zero surviving trapdoor-interface candidates and zero PKE finalists**. **Method** [Final]

No secure PKE, correctly directed reduction to EndRing, novelty result, implementation, parameter set, or impossibility theorem was established. All work was symbolic: zero scientific, implementation, formalizer, benchmark, or parameter-selection runs. The packets are scoped obstructions to their exact interfaces, not evidence that EndRing PKE cannot exist. [Final]

Method

The frozen method studies a basis-independent map

Trap_E : X_E -> Y_E

before another encryption wrapper is admitted. Public **Sample** and **Eval** must run from a bare supersingular curve on the exact EndRing challenge distribution. **Inv** must work from every complete evaluable basis of **End(E)**, including arbitrary **GL_4(Z)** presentation changes. Complete-ring necessity needs a non-factorization theorem or a reconstruction theorem from the smallest sufficient witness. The reduction direction is fixed as

selected-message attacker
-> **interface inverter or distinguisher**
-> **complete evaluable End(E) basis.**

An EndRing solver that can invert is an attack implication, not a security proof. Two mutually blind producers proposed distinct interfaces; the packets were snapshotted; a validator owned T1–T2, a red team owned T3–T4, and a third reviewer owned T5. Composition selected one structured gap for a fresh producer, whose packet received a second independent review. **Method** [Plan1] [Plan2]

Joint	Frozen question
T1	Are public sampling/evaluation exact and simulable from bare E , including distribution and abort mass?
T2	Is inversion typed, polynomial, and correct for every complete basis and basis change?
T3	Is the complete ring necessary, or does inversion factor through a smaller witness without reconstruction?
T4	Does an attacker yield a complete ring basis with explicit embedding, simulation, loss, runtime, and abort terms?
T5	Are encodings, malformed inputs, repetition, controls, sources, prior art, claims, and documentation complete?

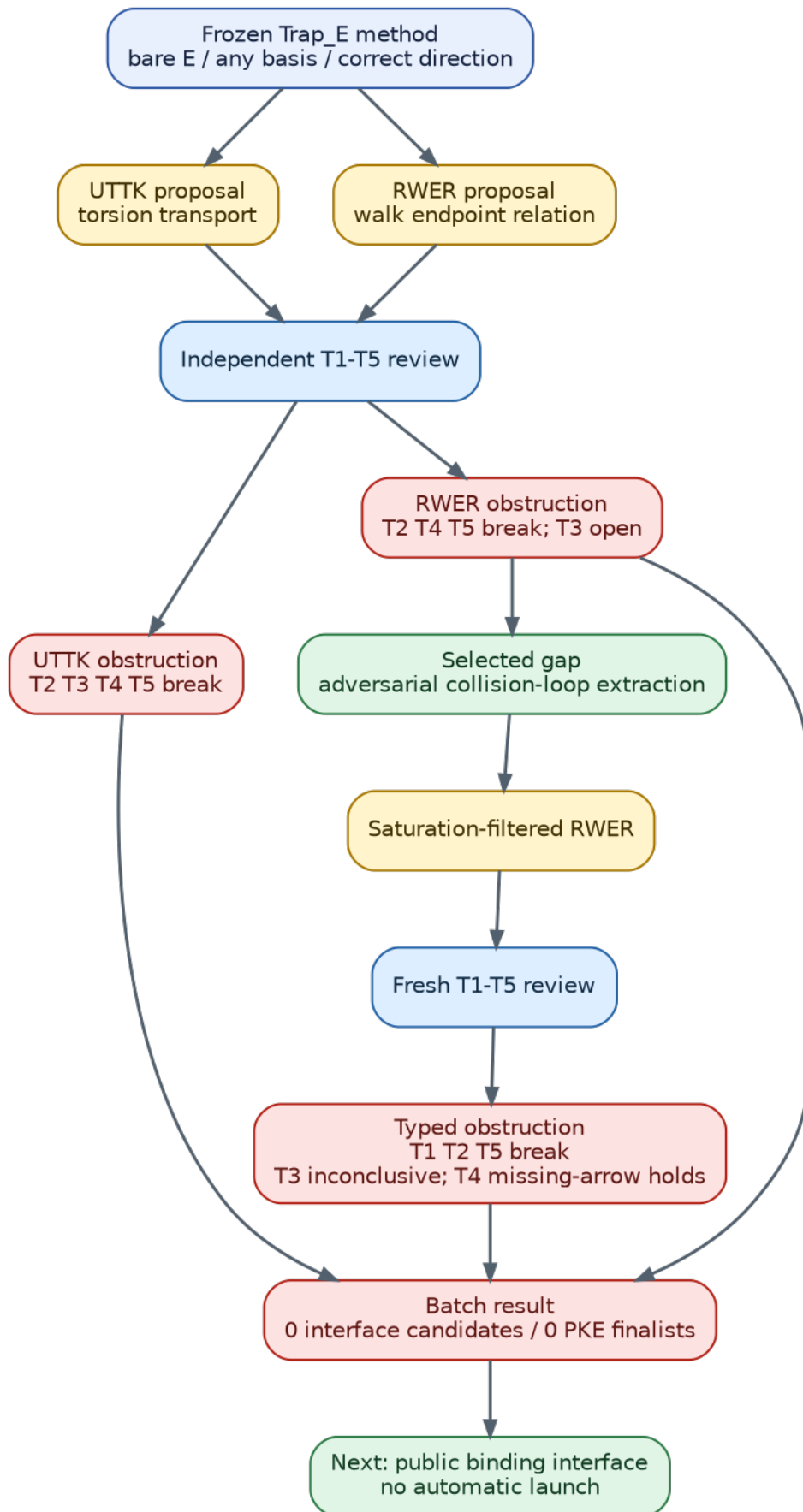


Figure 1: Candidate and review flow. Yellow nodes are proposals, blue nodes are independent review, red nodes are scoped failures, and green nodes are the selected gap/next² action. The diagram is navigation, not evidence.

Initial proposals and evaluation

UTTK: unoriented torsion-transport kernel interface

UTTK publicly samples a smooth cyclic kernel and a torsion basis, then emits the quotient curve and transported torsion points. Its intended inverse would compile any complete ring basis to one attack-suitable endomorphism `theta` and run a torsion-recovery procedure. The producer correctly returned a typed obstruction: the any-basis compiler is undefined, the recovery theorem is not verified at this scope, and the route factors through the smaller witness `theta`. Primary Seta text was not read; repository pointer scope suggests substantial overlap, so novelty is unverified. [A]

The validator found T1 inconclusive because the exact random-point law, rejection tails, and abort mass are unspecified, and T2 breaks because no $\text{GL}_4(\mathbb{Z})$ -invariant any-basis inverse exists. [V1] The red team found T3 and T4 break: the only inverse architecture is `B -> theta -> kernel`, and a bare-curve simulator that samples the kernel already knows what an inverter returns. No selected-message embedding or inverter-to-ring extractor exists. [R1] T5 breaks because adversarial transcript validity, codomain-isomorphism encoding, and repeated-use semantics are incomplete. [T51]

RWER-Trap: random-walk endpoint routing

RWER publicly samples an explicit small-degree isogeny walk and publishes a canonical endpoint relation. A desired trapdoor router would use any complete source ring basis to find a path to the endpoint. A different returned path could be composed with the sampled path to form a source endomorphism. [B]

T1 is inconclusive because the exact edge law, automorphism normalization, field/model encoding, and retry tails are incomplete. T2 breaks because no basis-independent route algorithm is supplied. [V1] T3 remains inconclusive: a target path is a smaller target-specific witness, but no polynomial reusable smaller witness was proved sufficient. T4 breaks because guaranteed edge/dual-edge collisions yield scalar loops, while a valid inverter may return the original, canonical, or padded path; success gives no probability of non-scalar rank growth or maximal-order saturation. [R1] T5 breaks on endpoint/tag/certificate grammar and repeated-use view. [T51]

Initial interface	T1	T2	T3	T4	T5	Disposition
UTTK	inconclusive	breaks	breaks	breaks	breaks	typed obstruction
RWER-Trap	inconclusive	breaks	inconclusive	breaks	breaks	typed obstruction; one gap selected

Gap-fed proposal and fresh evaluation

The initial composition selected one gap only: strengthen or replace the endpoint relation so that successful adversarial inversion quantitatively forces useful non-scalar independent loops and full-order saturation, without assuming a selector, endpoint ring, connecting ideal, orientation, or planted alignment. [C1] [Gap]

The fresh producer proposed saturation-filtered RWER. For sampled walks $\mathbf{x}$ and returned walks $\mathbf{x}'$, the reduction forms

```
alpha_i = dual(x'_i) o x_i in End(E).
```

It also wrote the minimal selected-message mask adapter ($\mathbf{y}, \mathbf{m} \text{ xor } \mathbf{h}(\mathbf{x})$). The packet identifies two real barriers: decryption through arbitrary valid inverses needs $\mathbf{h}$ to be fiber-constant, while predicting a fiber bit supplies no decision-to-search compiler; ordinary endpoint validity also contains no rank, discriminant, or maximal-order predicate. [C]

Fresh validation found the positive T1/T2 repair breaks. The sampler law, abort/cost accounting, and repeated view remain prose, while the universal any-basis `Inv` is explicitly missing. The artifact is coherent only as a typed obstruction. [V2]

Fresh red-team review preserved the T4 missing-arrow obstruction but corrected the producer's strongest T3 statement. The equation $\epsilon^q * \mathbf{u}_I$ is unjustified without an independence and repeated-call model. Moreover, "no

positive lower bound was derived” is not yet a proof that the concrete infimum is zero: a canonical or padded selector must be shown to preserve the exact fixed-length, degree-word, and endpoint rules. T3 is therefore **inconclusive**, not a proved zero-bound theorem. [R2]

T5 breaks because the exact byte grammar, equality role of degree words and tags, repeated-use adaptivity/state/randomness coupling, and composed abort semantics are unspecified. [T52]

Gap-fed interface	T1	T2	T3	T4	T5	Disposition
Saturation-filtered RWER	breaks	breaks	inconclusive	holds only as scoped missing-arrow obstruction	breaks	typed obstruction with reviewer correction

Authorized correction controls

This batch carried the earlier J4/J5 correction semantics as regression constraints rather than revoting the historical packet. The fresh T5 review verified the additive Milne metadata mapping against the frozen independent audit record: printed pages 69–70 map to PDF 74–75, printed 74 to PDF 79, and printed 75/Lemma 6.12 to PDF 80 for source bytes SHA-256 646c0c4f193cdaa35f32c7d60fbd46a75e2f5187db4301810e8613726e3ebbee. No fresh primary PDF read occurred, and historical J4/J5 verdicts are unchanged. [Correction] [T51]

The normalized $x(X)$, $P=0$, lower-order-point, and digit-loop controls were also preserved: weakened exact-order premises are controls rather than valid- input attacks, and no digit-loop output is inferred without tie-handling pseudocode. Neither new interface relies on an $x(X)$ -only collision. [Method](#)

Requirement-to-evidence table

Requirement	Outcome	Evidence
Two distinct initial interfaces	complete	[A], [B]
Immutable initial snapshot	complete	[S1]
Independent initial T1–T5 review	complete	[V1], [R1], [T51]
One structured gap-fed proposal	complete	[Gap], [C]
Immutable gap-fed snapshot	complete	[S2]
Fresh revision T1–T5 review	complete	[V2], [R2], [T52]
Corrected locator/control carry-forward	complete at metadata/regression scope	[Correction], [T51], [T52]
Secure EndRing PKE or surviving interface	not established	[Final]

Exactly one next action

Before another PKE wrapper or RWER variant, define and audit a publicly checkable binding relation whose honest output has a unique or publicly equivalent preimage, while **Sample/Eval** use only bare **E** and **Inv** works from every complete evaluable ring basis. The exact parser and repeated-use game must precede the reduction. The binding predicate may not hide a canonical selector, endpoint order, connecting ideal, orientation, or planted basis. This successor is not launched automatically. [Final]

Source index

- [Method](#) [Frozen methodology](#)
- [Plan1] [Initial review plan](#)
- [Plan2] [Revision review plan](#)

- [A] [UTTK packet](#)
- [B] [RWER-Trap packet](#)
- [S1] [Initial snapshot](#)
- [V1] [Initial T1–T2 validation](#)
- [R1] [Initial T3–T4 red team](#)
- [T51] [Initial T5 review](#)
- [C1] [Initial composition](#)
- [Gap] [Structured gap](#)
- [C] [Gap-fed packet](#)
- [S2] [Gap-fed snapshot](#)
- [V2] [Fresh T1–T2 validation](#)
- [R2] [Fresh T3–T4 red team](#)
- [T52] [Fresh T5 review](#)
- [Correction] [Additive locator correction](#)
- [Final] [Final composition](#)